

SMART TRASH ALERTING SYSTEM USING IOT

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(Accredited by NAAC with 'A+' Grade)

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Department of Electronics and Communication Engineering

2024 - 2025

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Industrial Oriented Mini Project Report

Submitted in partial fulfilment

As a part of academics

in

ELECTRONICS AND COMMUNICATION ENGINEERING BY

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CERTIFICATE

Date: /06/2025

This is to certify that the project work entitled **SMART TRASH ALERTING SYSTEM USING IOT** is a bonafide work carried out by **R.KAVYA SRI, K.HRRUSHIKESH, A.KIRAN** bearing Roll Nos. **23JJ5A0407, 23JJ5A0403, and 22JJ1A0403** in partial fulfillment of the requirements as a part of Academics in **ELECTRONICS & COMMUNICATION ENGINEERING** by the Jawaharlal Nehru Technological University Hyderabad during the academic year 2024-25. The results embodied in this report have not been submitted to any other University or Institution for the award of any degree or diploma.

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ACKNOWLEDGEMENT

The success in this project would not have been possible without timely help and guidance by many people. We wish to express our sincere gratitude to all those who have helped and guided us for the completion of the project.

It is our pleasure to thank our **Project Guide Dr. B. PRABHAKAR, Professor and Head of ECE, JNTUH UCEJ** for the guidance and suggestions throughout the project, without which we would not have been able to complete this project successfully.

We wish to express our sincere gratitude to our **Project Coordinator Smt. M. TIRUPATHAMMA, Associate Professor, ECE, JNTUH UCEJ**, for whole hearted support and encouragement.

We express our sincere gratitude to **Dr. S. VISWANADHA RAJU, Principal** and **Dr. K. SRINIVAS, Vice-Principal, JNTUH UCEJ**, for their encouragement and providing facilities to accomplish our project successfully.

We would like to thank our classmates, all faculty members and non-teaching staff of ECE Department, JNTUH UCEJ for their direct and indirect help during the project work. Finally we wish to thank our family members and our friends for their interest and assistance that has enabled to complete the project work successfully.

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ABSTRACT

Modern urban areas demands real-time monitoring and efficient communication to address challenges such as delayed pickups, overflowing bins, inefficient collection schedules, and poor communication between waste handlers. Traffic congestion further delays timely waste removal, while limited municipal resources and lack of workers makes it hard to manage things well. The decaying waste produces unpleasant odors that make nearby areas uncomfortable for residents and visitors, which can spread various diseases and pose health risks.

These issues highlight the urgent need for smart, technology-driven solutions to improve urban sanitation and sustainability. In this project, it not only tracks waste levels and bin locations but also sends automated message alerts to waste collectors or municipal staff for timely intervention. The system uses ultrasonic sensors to detect the level of waste inside bins. A microcontroller (e.g., NodeMCU Wi-Fi module) processes this data and transmits it to a cloud server, which processes bin status. When bins reach predefined threshold levels, the system sends a real-time notification to the user's smartphone via the **Blynk app using Wi-Fi** connectivity or third-party automation services.

Overall, this project highlights the potential of IoT technology to address real-world problems and contributes to the advancement of smart city infrastructure by offering a scalable, cost-effective, and environmentally responsible solution. This results integration of IoT and real-time-based communication bridges the gap between data collection and human response, ensuring that waste collection is faster, smarter, and more responsive. This project is helpful for government '**SWATCHA BHARAT ABHIYAN**'.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

In today's fast-developing world, urbanization and industrialization have drastically changed the way communities function. With increasing population density, residential complexes, commercial zones, and public areas are generating large volumes of waste on a daily basis. As cities expand, the pressure on sanitation systems grows, making it difficult to maintain clean and healthy surroundings through traditional methods. One of the major challenges observed is the lack of real-time monitoring and timely removal of garbage, which often results in bins becoming overloaded before any action is taken.

Overflowing bins are not only unpleasant to look at but also contribute to the spread of foul odors, germs, and harmful bacteria. This creates an unhygienic environment, increasing the risk of disease transmission and affecting public well-being. Additionally, such situations attract insects and stray animals, further worsening the conditions.

Technological advancements have opened the door to intelligent systems that can automatically monitor the status of bins and provide timely updates. Such innovations also support the broader vision of smart cities by enabling automation, reducing human effort, and improving the overall quality of urban life. Encouraging the use of digital solutions in sanitation not only helps maintain a cleaner community but also promotes responsible practices and environmental awareness.

1.2 AIM OF THE PROJECT

The aim of this project is to design and implement an IoT-based smart alerting system that can monitor the fill level of garbage bins in real-time using ultrasonic sensors and notify users through a mobile application interface. The system seeks to automate the monitoring process, reduce manual inspection, and ensure timely removal of garbage to maintain hygiene and cleanliness in public and private spaces.

1.3 MAIN OBJECTIVES OF THE PROJECT

The main objectives of project are:

1. To study the concept of a smart trash alerting system using IoT technology for efficient bin monitoring and management.
2. To select and integrate suitable hardware and software components for accurate sensing and real-time data transmission.
3. To develop an automated alert mechanism that notifies users when the trash bin reaches a predefined fill level.
4. To ensure timely response and reduce the chances of overflow, thereby contributing to a cleaner and more hygienic environment.

1.4 METHODOLOGY OF THE PROJECT

The methodology outlines the systematic process followed to design, develop, and implement the smart trash alerting system using IoT technology. The approach is divided into several key stages as follows:

1. Problem Identification

- Identified the need for a system to monitor trash bin status in real time.
- Observed challenges in traditional garbage removal methods, including lack of timely response and resource inefficiency.

2. Requirement Analysis

- The system requires accurate bin-level detection, real-time alerting, and wireless communication using IoT.
- It must be low-cost, power-efficient, easy to install, and scalable for wider use.

3. Hardware Selection and Integration

- Choose essential components such as:
 - NodeMCU ESP8266 (for Wi-Fi communication and control)
 - Ultrasonic Sensor (HC-SR04) (for measuring bin fill level)
 - Power source, wires, and connectors
- Integrated components on a breadboard or fixed circuit setup.

4. Software Development

- Programmed the NodeMCU using Arduino IDE.
- Wrote code to measure distance using the ultrasonic sensor and calculate the garbage level.
- Connected the NodeMCU to the Blynk App via Wi-Fi for real-time alert notifications.

5. System Testing and Calibration

- Tested the sensor readings under various bin conditions (empty, half-full, full).
- Calibrated the sensor distance threshold for alert activation.

6. Deployment and Observation

- Placed the system inside a garbage bin prototype.
- Monitored performance, alert accuracy, and real-time response via the mobile application.

This step-by-step methodology ensured a structured development process, leading to a working prototype capable of monitoring and alerting users when the trash bin reaches its capacity.

1.5 SIGNIFICANCE OF THE PROJECT

The Smart Trash Alerting System using IoT plays a vital role in addressing the issues caused by unattended and overflowing trash bins in public spaces. By providing timely alerts when the bin reaches a certain capacity, the system ensures cleanliness and hygiene in the surrounding areas. It reduces the dependency on manual monitoring and enables efficient use of manpower. Through IoT integration, real-time status updates are delivered to the user, supporting smarter and more responsive urban management. This solution aligns with the goals of smart city development and environmental sustainability. Additionally, it helps minimize health risks associated with waste overflow. This project contributes to cleaner surroundings and better public sanitation practices. By providing data for analysis, it enables better planning and management of waste infrastructure. Overall, the project is a cost-effective, scalable, and eco-friendly solution that contributes to sustainable urban development.

1.5 ORGANIZATION OF THE THESIS

This entire thesis is divided into five chapters including introductions and conclusions. This report explains Smart trash alerting system using IOT. The organization of thesis is explained as here with

Chapter 1: Describes the overview of the project which explains introduction, aim, objectives, significance and thesis of the project.

Chapter 2: Presents literature review of the project which gives overview of various existing techniques.

Chapter 3: Explains about the hardware and software description of the project.

Chapter 4: Explains about the hardware implementation, flowchart, block diagram and working operation of the project.

Chapter 5: Gives a discussion on the experimental results generated.

Chapter 6: Conclusion, future scope and references of the project.

1.7 CONCLUSION

In the introduction chapter highlights the growing need for smart solutions in urban cleanliness and environmental management. The primary aim of this project is to develop a system that monitors bin fill levels and sends real-time alerts for timely collection. Key objectives include automation, cost-efficiency, and improved hygiene. The system plays a crucial role in reducing manual checks, preventing overflow, and supporting clean surroundings. The methodology involves using sensors, NodeMCU, and cloud platforms for data transmission and monitoring. The thesis is organized into chapters covering literature review, system design, implementation, testing, and analysis. Overall, this chapter lays the foundation for building an intelligent, IoT-based alerting system.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

The rapid urbanization and population growth have significantly increased waste generation, posing challenges to efficient waste management in cities and communities. Traditional waste management systems often lack real-time monitoring, leading to issues such as overflowing trash bins, inefficient collection schedules, and environmental pollution. To address these problems, the integration of the Internet of Things (IoT) into waste management has emerged as a promising solution. This literature survey explores existing research and developments in the domain of IoT-based smart trash alerting systems.

Additionally, it reviews the methodologies for data collection, processing, and alert generation, as well as the integration of cloud platforms and mobile applications for real-time monitoring. The survey also highlights challenges such as power management, scalability, and cost-effectiveness, while identifying gaps in current research and potential areas for future exploration. By analyzing prior work, this study aims to provide a comprehensive understanding of the advancements and limitations in the field, laying the foundation for developing an efficient and scalable smart trash alerting system using IoT.

2.2 LITERATURE SURVEY

Several research studies and innovations have been proposed in recent years to tackle the growing concern of urban waste management through smart technologies. The integration of the Internet of Things (IoT) in waste monitoring has gained attention due to its potential to improve efficiency, reduce costs, and support sustainability.

1. S. Longhi et al. (2012) proposed a smart waste management system based on RFID, GPS, and GPRS technologies. Their system tracked garbage trucks and bin locations, aiming to optimize the collection routes and schedules. However, it lacked real-time bin level detection.

2. Narayan Sharma et al. (2015) introduced a system using ultrasonic sensors to detect waste levels in bins and send alerts via GSM modules. While effective in monitoring, the use of GSM increased operational costs and power consumption.

3. V. Sakshi et al. (2016) developed a system using Arduino and ultrasonic sensors for local alerts. The limitation was its inability to send notifications remotely or store data for analysis, which restricted its use in smart city applications.

4.A. Joshi and K. Patel (2018) used NodeMCU and Wi-Fi modules to connect bins to cloud platforms like Blynk. This allowed real-time notifications and dashboard visualization. Their work laid the foundation for low-cost, scalable solutions using IoT and open-source platforms.

5.R. Sivaranjani et al. (2020) enhanced smart bin systems with waste segregation features using multiple sensors and AI-based classification. Although advanced, the system required complex implementation and higher investment.

6. V. Sakshi et al. (2016) designed a basic smart bin using Arduino and ultrasonic sensors, with LED indicators for alerts. It lacked wireless or mobile communication features. The system was best suited for small indoor applications.

7. Dr. S. Vinoth Kumar et al. (2017) implemented an IoT-based smart trash alert system using NodeMCU and Wi-Fi. It sent real-time alerts to a mobile app. The project was cost-effective and scalable for smart cities.

8. R. Mamatha et al. (2018) created a smart bin using ESP8266 and Firebase to monitor garbage levels. It stored and displayed data on the cloud. However, it required a stable internet connection for effective operation.

9. S. Jain and R. Bagrodia (2019) built a system using NodeMCU and the Blynk app to alert users via mobile notifications. The bin level was also shown on a real-time dashboard. The setup was simple and user-friendly.

10. P. Shetty et al. (2020) proposed a system for waste level detection and segregation of wet and dry waste. It used multiple sensors to improve environmental impact. While effective, the system needed complex hardware.

11. M. Priya and S. Divya (2021) developed a solar-powered smart bin with gas, ultrasonic, and moisture sensors. It offered advanced features for smart monitoring. The system was more suitable for large-scale or industrial use.

12. R. Thakker and R. Narayanamoorthi (2016) built a Wi-Fi-based monitoring system using IR sensors. Bin data was displayed on a central server for analysis. It was designed for municipality-level use but lacked mobile alerts.

13. S. Raut and S. Bodkhe (2017) used NodeMCU with Thingspeak for bin level monitoring and cloud visualization. Their system supported real-time data logging. However, mobile notification was not included.

14. M. Ismail and N. A. Ali (2020) created a smart bin powered by solar energy with multi-sensor integration. It measured gas emissions, moisture, and garbage level. The project was rich in features but had higher cost.

From the above studies, it is evident that while several models exist, many systems face challenges such as high cost, limited scalability, and lack of real-time data visualization. Our proposed system overcomes these limitations by using NodeMCU with ultrasonic sensors and the Blynk app for real-time bin status monitoring and alerting, making it a practical and cost-effective solution for modern waste management.

2.3 CONCLUSION

In this chapter we have discussed about the various studies related to the project in literature review. The literature review has explored various existing systems related to smart bin monitoring and their technological approaches. While many systems offer partial automation, most lack real-time alerting, scalability, and cost-efficiency. Recent developments in IoT have enhanced monitoring capabilities but still present challenges like integration and power consumption. Through this review, several research gaps have been identified. These gaps highlight the need for a more reliable, efficient, and low-cost system. This understanding forms the foundation for the proposed smart alerting system in this project.

CHAPTER 3

HARDWARE AND SOFTWARE DESCRIPTION

3.1 INTRODUCTION

This chapter provides a brief description of the hardware tools used for the development of the project. This presents a detailed overview of the hardware and software components used in the development of the Smart Trash Alerting System using IoT. The system relies on a combination of electronic components and programming tools to detect bin fill levels and send alerts. The hardware section includes essential components such as the NodeMCU microcontroller, ultrasonic sensor, power supply, and connecting wires. The software section describes the tools and platforms used for coding, data transmission, and real-time monitoring—such as the Arduino IDE and IoT platforms like Blynk or ThingSpeak. Together, these components work seamlessly to achieve the goal of an efficient and automated alerting system.

3.2 HARDWARE DESCRIPTION

1. NODEMCU (ESP8266)

NodeMCU is an open-source IoT platform based on the ESP8266 Wi-Fi module developed by Espressif Systems. It combines a microcontroller with built-in Wi-Fi capabilities, allowing easy and cost-effective development of wireless networking applications. NodeMCU is popular for creating smart devices and home automation systems due to its small size, low power consumption, and ease of use. It supports programming in multiple languages such as Lua, Arduino C/C++, and MicroPython, making it accessible for beginners and advanced users alike. The board includes a USB interface for easy programming and power supply, and it can be directly connected to a computer using a micro-USB cable. With support for digital and analog I/O, UART, SPI, I2C, PWM, and ADC, it enables versatile connectivity with sensors and modules. NodeMCU is widely used in projects like smart trash systems, weather monitoring, smart lighting, and other IoT-based solutions.

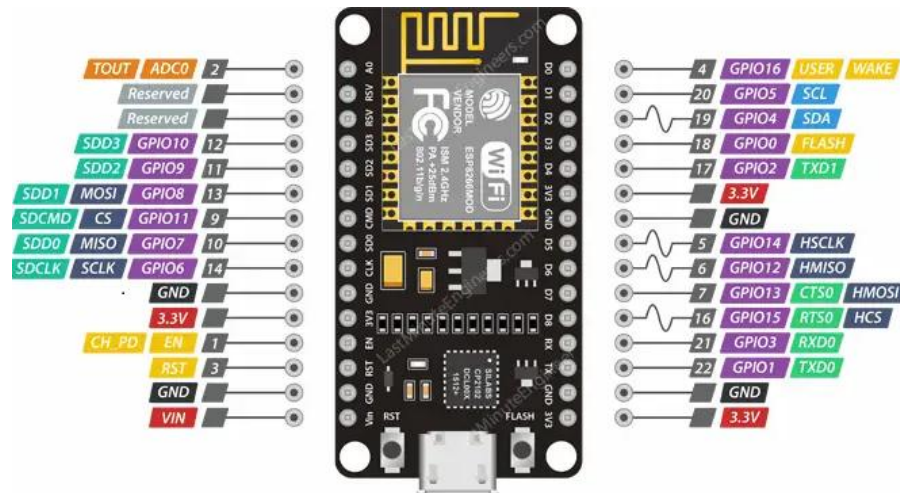


Figure 3.1 : Pin description of NodeMCU (ESP8266)

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NodeMCU ESP8266 Wi-Fi module specifications :

Microcontroller	ESP-8266 32-bit
NodeMCU Model	NodeMCU v1.0 (Amica)
Clock Speed	80-160 MHz
USB to Serial	CP2102 (Silicon Labs)
Operating Voltage	3.3V
Input Voltage	4.5V-10V
Flash Memory/SRAM	4 MB / 128 KB
GPIO Pins	17
Digital I/O Pins	11
Analog In Pins	1
PWM Pins	All GPIOs (Software PWM)
WiFi Built-In	802.11 b/g/n (2.4 GHz)
Temperature Range	-40°C to +125°C (chip) / ~ -20°C to +85°C (board)

Table 3.1 : Specifications of NodeMCU (ESP8266)

Each pin is described briefly below:

1. GPIO0 (D3)

GPIO0 is a multifunctional pin used during boot to select flash mode. It must be pulled HIGH during normal operation; pulling it LOW while resetting puts the board into firmware flash mode. It's usable as a digital I/O but should be handled carefully. Avoid using it for output devices that pull it LOW during boot.

2. GPIO1 (TX / D10)

GPIO1 is the UART TX (transmit) pin used for serial communication with the computer or other devices. It is active during flashing and debugging. While it can be used as a general-purpose digital output, it may interfere with USB communication. It's best used for serial tasks or left unused during normal operation.

3. GPIO2 (D4)

GPIO2 is a digital I/O pin and also connected to the onboard LED. It must be HIGH during boot to start correctly. It's commonly used for digital output tasks like blinking the LED. Be cautious not to pull it LOW at boot time.

4. GPIO3 (RX / D9)

GPIO3 serves as the UART RX (receive) pin, used to receive serial data from the computer or other serial devices. Like GPIO1, it's best used for serial communication. It can function as a digital input, but using it may disrupt USB programming or logging.

5. GPIO4 (D2)

GPIO4 is a regular digital I/O pin and can be safely used for most functions. It is commonly used for I2C communication as the SDA line. It supports PWM, I2C, and other general purposes. It's stable during boot and has no special boot restrictions.

6. GPIO5 (D1)

GPIO5 is another safe digital I/O pin and is often used as the I2C SCL line. It supports PWM, SPI, and other digital functions. It's reliable at boot and does not affect startup mode. It is often used alongside GPIO4 in sensor connections.

7. GPIO12 (D6)

GPIO12 is a general-purpose digital I/O pin and part of the SPI interface (MISO). It can be used for digital input, output, PWM, and other standard functions. Avoid pulling it HIGH or LOW during boot if you are unsure, though it's generally safe. It works well for interfacing with devices like sensors or LEDs.

8. GPIO13 (D7)

GPIO13 is commonly used as the SPI MOSI pin. It's a general-purpose I/O pin that also supports PWM. This pin can be used for various digital applications. It's stable at boot and doesn't interfere with the boot process.

9. GPIO14 (D5)

GPIO14 is often used as the SPI clock (SCLK) pin and is a general digital I/O. It supports all functions including PWM and SPI. This pin is safe to use and doesn't affect booting. It's suitable for high-speed communications or general outputs.

10. GPIO15 (D8)

GPIO15 must be kept LOW during boot for the ESP8266 to start correctly. It is a digital I/O pin and part of the SPI CS (chip select) line. Avoid using pull-ups or devices that drive this pin HIGH at startup. Once booted, it can be used normally.

11. GPIO16 (D0)

GPIO16 is a special pin used for waking the chip from deep sleep. It does not support PWM or I2C but works as a basic digital I/O. It's ideal for simple tasks like triggering interrupts or status LEDs. It's independent of boot configuration.

12. A0 (ADC0)

A0 is the only analog input pin on the NodeMCU. It can measure voltages from 0 to 1V only, so a voltage divider is required for higher voltages. It's used to read analog sensors like LDRs or temperature sensors. Only one analog channel is available on the board.

Power Pins Overview :

13. VIN

The VIN pin is used to supply input power to the board when not using USB. You can provide 5V directly to this pin from a battery or adapter. It powers the onboard voltage regulator which provides 3.3V to the ESP8266. It should not exceed 6V.

14. 3V3

This pin provides a regulated 3.3V output from the onboard regulator. It can be used to power external 3.3V sensors and devices. Do not input voltage into this pin. It supplies limited current (approx. 500mA max) depending on the board design.

15. GND

GND is the ground pin and must be connected to the ground of your circuit. All electronic components and modules connected to NodeMCU should share this ground. Multiple GND pins may be present for easier wiring.

Control Pins :

16. EN (Enable)

The EN pin enables or disables the ESP8266 chip. It must be pulled HIGH for the module to run. Pulling it LOW turns off the chip and stops execution. It is usually tied HIGH through an onboard resistor.

17. RST (Reset)

RST is used to reset the NodeMCU manually. Pulling it LOW restarts the board and re-runs the code from the beginning. It can be connected to a push button for manual reset purposes.

Applications of NodeMCU (ESP8266)

1. Home Automation Systems – Control lights, fans, and appliances over Wi-Fi.
2. IoT-Based Projects – Real-time monitoring and control using internet connectivity.
3. Smart Agriculture – Soil moisture monitoring, irrigation control, and weather tracking.
4. Smart Trash Monitoring – Detect bin levels and send alerts for timely waste disposal.
5. Wireless Sensor Networks – Collect and transmit environmental data like temperature, humidity, etc.
6. Home Security Systems – Motion detection, door sensors, and surveillance control

2. ULTRASONIC SENSOR

The HC-SR04 ultrasonic sensor is a widely used distance measuring device in the field of electronics and embedded systems. It operates by using ultrasonic sound waves, which are inaudible to human ears, to determine the distance of an object from the sensor. The sensor consists of two cylindrical transducers: a transmitter and a receiver. When powered, the transmitter emits a short burst of 40 kHz sound waves, which travel through the air until they encounter an object. These waves then bounce back to the sensor and are detected by the receiver. The time taken for this echo to return is measured, and using the known speed of sound in air (approximately 343 m/s), the distance to the object is calculated using the time-of-flight method.

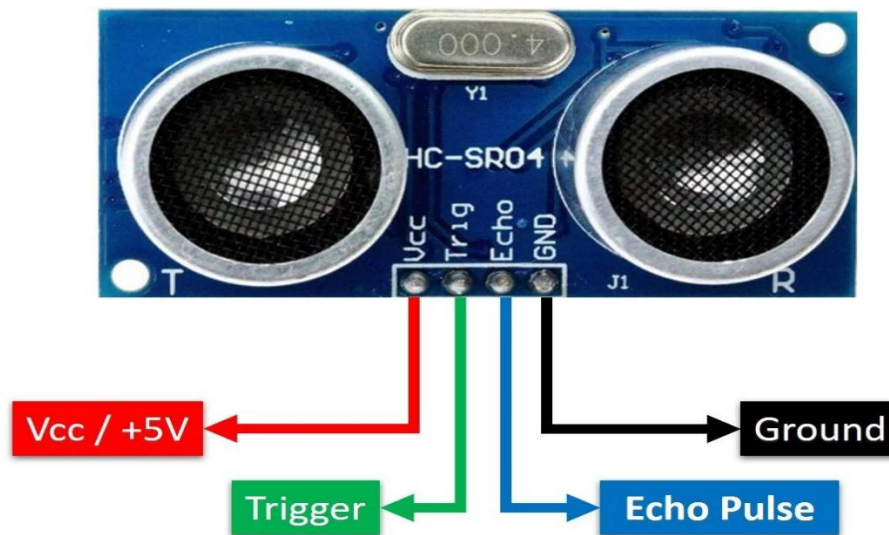


Figure 3.2 : Ultrasonic Sensor

Vcc and **Ground** are used to power sensor. We should supply 5 volts to the Vcc pin and connect the GND pin with the ground terminal of the power supply.

Trigger: It is an input pin. A trigger pin is used to initiate the ultrasonic sensor to start distance measurement or distance ranging. When users want to get distance measurements from the sensor, we apply a 10 μ s pulse to this pin.

Echo: This is a pulse output pin. The echo pin produces a pulse as an output. The width of pulse or on-time of the pulse depends on the distance between the ultrasonic sensor and the obstacle which is placed in front of the HC-SR04 sensor. In idle conditions, this pin remains at an active low level.

How HC-SR04 Sensor Works?

HC-SR04 ultrasonic sensor measures distance by using inaudible ultrasonic sound waves of 40KHz frequency. Like sound waves, ultrasonic waves travel through the air and if there is any obstacle in front of them, they reflect according to their angle of incidence. Moreover, if an object is placed parallel to an ultrasonic transmitter, ultrasonic waves reflect exactly at an angle of 180 degrees. Therefore, for distance measurement with HC-SR05 sensor, we place the object under test exactly in a parallel position with an ultrasonic sensor as shown in the figure below.

HC-SR05 ultrasonic sensor consists of two basic modules such as ultrasonic transmitter and ultrasonic receiver module. The transmitter circuit converts an electrical signal into a 40KHz burst of 8 sonar wave pulses. The input electrical signal to the transmitter circuit is 10 μ s pulse input to the trigger pin of the HC-SR04 sensor. As we mentioned earlier, we apply this trigger input signal through ESP8266 NodeMCU or any microcontroller. On the other hand, the ultrasonic receiver circuit listens to these ultrasonic waves which are produced by the transmitter circuit.

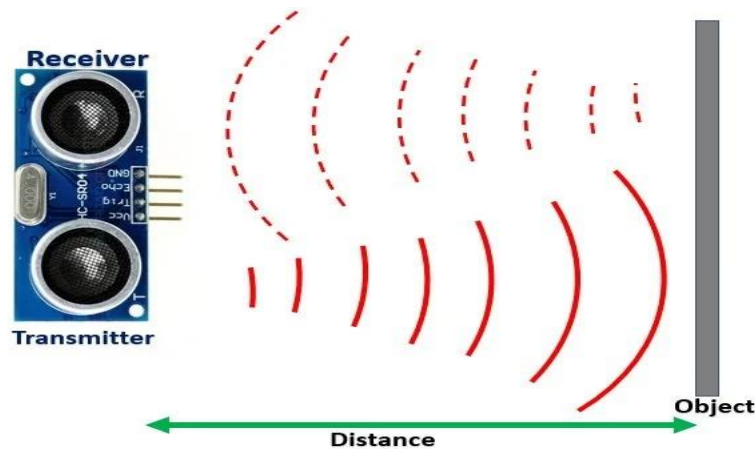


Figure 3.3 : Object under test in parallel position with an ultrasonic sensor

1. To start ranging with HC-SR04, first, we apply 10 μ s pulse to the trigger pin of the HC-SR04 sensor from the ESP8266 NodeMCU digital output pin.
2. As soon as 10 μ s input trigger signal becomes active low, the transmitter circuit produces a burst of 8 ultrasonic sonar pulses. At the same time, the Echo pin also makes a transition from a logic low level to a logic high level.
3. When the Echo pin goes high, We start to measure time with the ESP8266 NodeMCU duration measurement function.

4. These waves travel through the air and if there is any object placed in parallel to the sensor, these waves reflect back after a collision with the object.
5. As soon as the ultrasonic waves received by the receiver circuit after striking with an object, the echo pin goes low. ESP8266 NodeMCU detects this transition of echo output signal from active high to an active low level and stops the measurement.

In short, by measuring the on-time of the Echo output pulse signal, we can measure the distance. The following figure illustrates the echo output signal with respect input trigger signal and 8 sonar pulses.

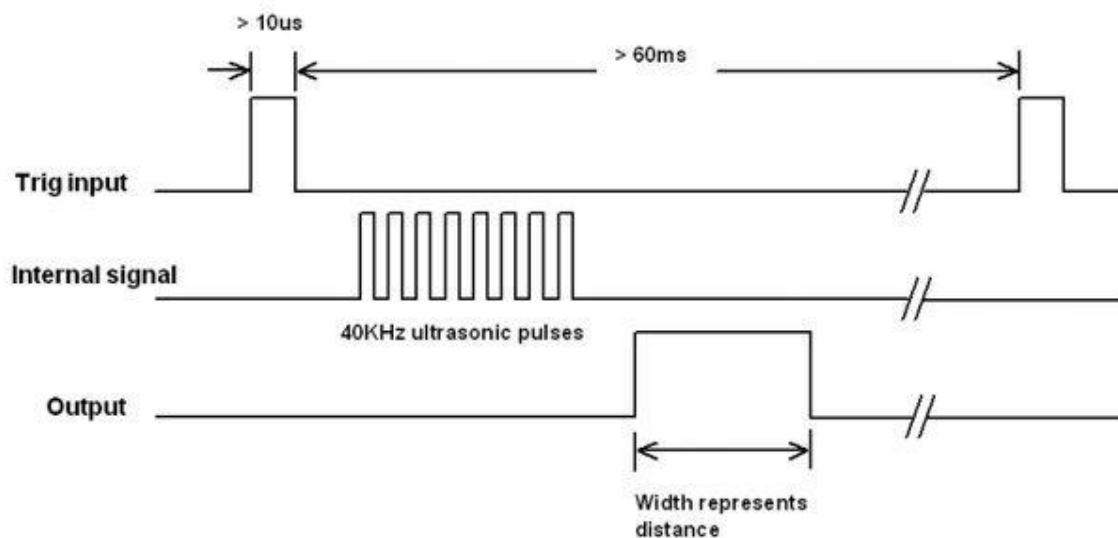


Figure 3.4 : Echo output signal with respect input trigger signal

The duration for which the echo output signal remains high depends on the distance between the ultrasonic sensor and the object which we place in front of the sensor. Higher is the distance, the higher the time sonar waves will take to reach back to the ultrasonic receiver circuit. Because ultrasonic waves travel through the air with the speed of sound and speed remains constant.

Specifications of Ultrasonic Sensor (HC-SR04)

- Operating Voltage: 5V DC
- Operating Current: 15 mA
- Ultrasonic Frequency: 40 kHz
- Measuring Distance Range: 2 cm to 400 cm (0.02 m to 4 m)

- Measurement Accuracy: ± 3 mm
- Measuring Angle: 15 degrees
- Trigger Pulse Width: 10 μ s
- Echo Pulse Output: Time in microseconds
- Response Time: < 50 ms
- Sensor Dimensions: Approximately 45 mm $\times$ 20 mm $\times$ 15 mm
- Communication Interface: Digital (Trigger and Echo Pins)
- Compatible Microcontrollers: Arduino, NodeMCU, ESP32, Raspberry Pi, etc.
- Working Temperature Range: -15°C to +70°C
- Weight: ~10 grams

Applications of Ultrasonic Sensor (HC-SR04)

1. Obstacle Detection in Robots – Helps robots avoid collisions.
2. Smart Trash Bins – Detects the fill level and sends alerts.
3. Distance Measurement Tools – Used in digital measuring devices.

3.3 SOFTWARE DESCRIPTION

1. Arduino IDE (Programming Platform)

The Arduino Integrated Development Environment (IDE) is an open-source platform used to write, compile, and upload code to microcontroller boards like NodeMCU (ESP8266). It supports Embedded C programming and a wide range of libraries for sensors, displays, and communication modules.

In this project:

- The Arduino IDE is used to write the logic for reading the ultrasonic sensor data, processing it, and sending it to the Blynk Cloud via Wi-Fi.
- It allows easy integration of libraries like ESP8266WiFi.h, BlynkSimpleEsp8266.h, and NewPing.h.
- It helps in defining and testing different bin fill levels and response logic like triggering alerts when the trash is full.

2. Embedded C (Programming Language)

Embedded C is a specialized version of the C language used to program microcontrollers and embedded systems. It provides direct access to hardware with minimal system overhead, making it ideal for real-time control applications.

In this smart bin system:

- The code is written in Embedded C to configure the trigger and echo pins of the ultrasonic sensor and measure distance accurately.
- Based on the measured distance, the trash level percentage is calculated.
- If the trash level exceeds a certain threshold (e.g., 80% full), the NodeMCU triggers a message to the user.
- Embedded C also controls the timing of readings, handles Wi-Fi reconnection, and ensures that data is pushed to the Blynk app at regular intervals.

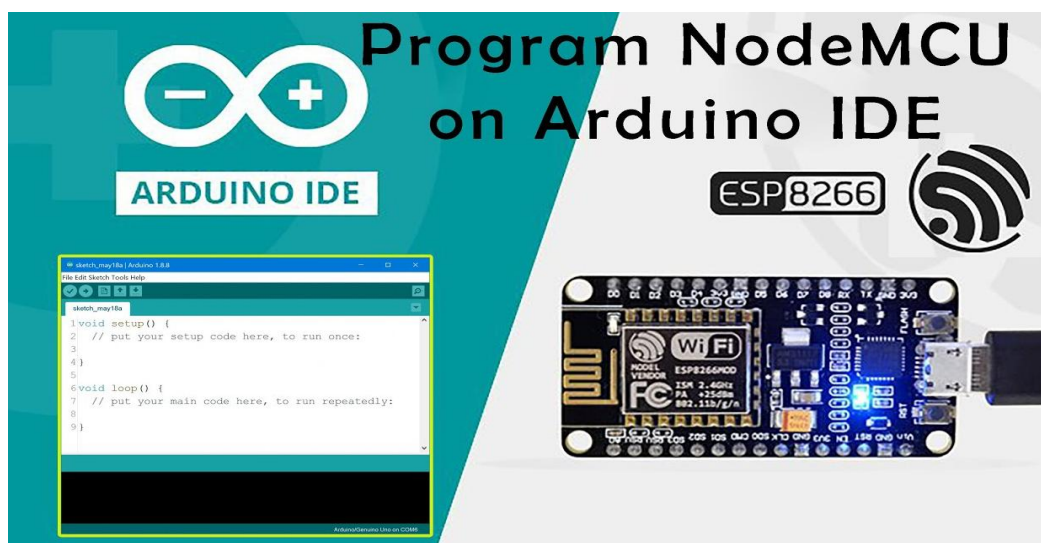


Figure 3.5 : Arduino IDE

3. Blynk App (Frontend User Interface)

The Blynk App acts as the frontend interface that communicates with the hardware over the internet. It is used to remotely monitor the status of the dustbin and receive alerts.

Key features implemented in the Blynk app:

- Gauge Widget or Level Bar to visually show how full the bin is (in %).

- Notification Widget to alert the user when the bin is full.
- SuperChart Widget (optional) to display trends in bin usage over time.
- Real-time data synchronization with the NodeMCU using the Blynk Cloud.

How it works:

- The NodeMCU sends the fill-level data to the Blynk server using your Blynk authentication token.
- The app fetches the data and updates the widgets on your mobile screen.
- If a certain threshold is reached (e.g., trash level $\geq 90\%$), the app sends a push notification to alert the user.

3.4 CONCLUSION

In this chapter we have discussed about the hardware tools like NodeMCU (ESP8266), Ultrasonic sensor and their applications used for the implementation of the project. This chapter provided a detailed overview of the hardware and software components used in the Smart Trash Alerting System. The NodeMCU, ultrasonic sensor, and power supply form the core of the hardware setup, enabling accurate level detection. On the software side, tools like Arduino IDE and Blynk were used for coding, monitoring, and data transmission. Each component's function and specifications were discussed to ensure proper system integration. This chapter builds the technical foundation necessary for the implementation phase ahead.

CHAPTER 4

IMPLEMENTATION

4.1 INTRODUCTION

In this chapter we will discuss about block diagram, flow chart and hardware implementation of the project. This chapter presents the practical implementation of the Smart Trash Alerting System using IoT. It begins with the block diagram, which outlines the system's overall structure and key components. The flow chart is then used to explain the logical sequence and decision-making steps involved in the operation. The hardware implementation section describes how the components are physically connected and configured. Following this, the working operation explains how the system detects bin levels and sends alerts in real-time. This chapter demonstrates how the proposed design is successfully put into action.

4.2 BLOCK DIAGRAM

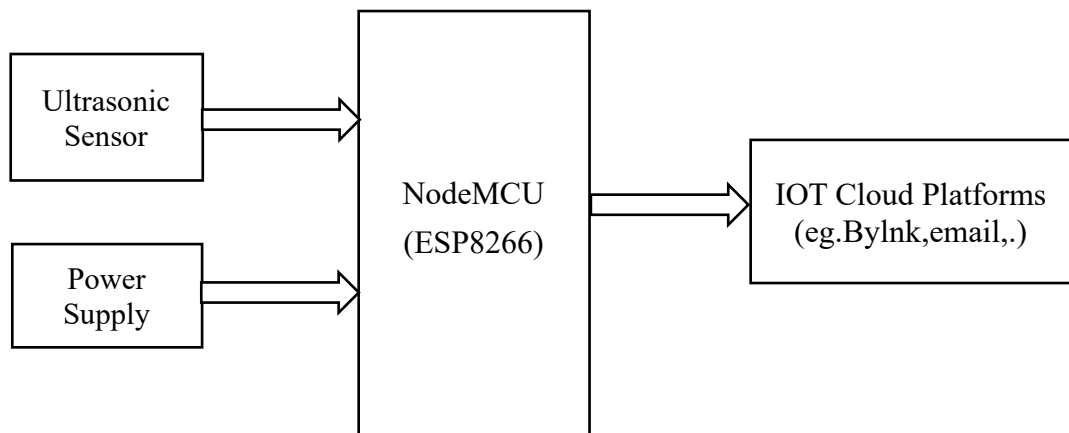


Figure 4.1 : Block diagram of Smart Trash Alerting System using IOT

Figure 4.1 illustrates the block diagram of the real-time IOT based smart trash alerting using which consisting of Power Supply, Ultrasonic Sensor, NodeMCU (ESP8266), Wi-Fi Module, Blynk Cloud, Blynk Mobile App, and an optional Buzzer or LED Indicator. The power supply provides the required voltage to all components. The ultrasonic sensor detects the trash level, and the NodeMCU processes this data. Using its built-in Wi-Fi module, the NodeMCU sends

the data to the Blynk cloud. The Blynk mobile app receives and displays the status, while the buzzer or LED provides local alerts when needed.

4.3 FLOW CHART

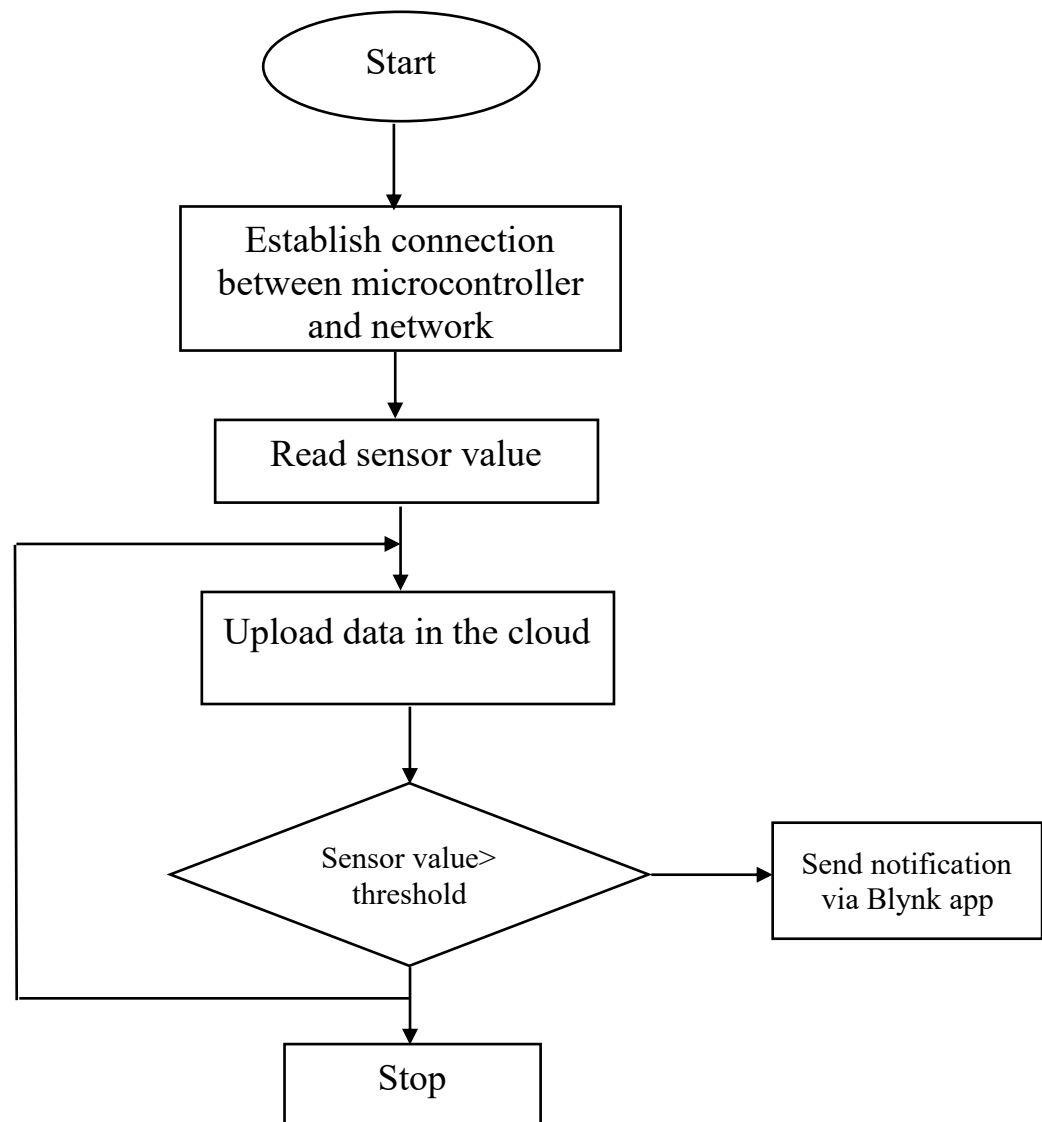


Figure 4.2 : Flow chart of the System

Figure 4.2 represents the flow chart of working process of the Smart Trash Alerting System. It starts by establishing a connection between the microcontroller (NodeMCU) and the network. Once connected, the ultrasonic sensor reads the trash level and uploads the data to the cloud. If the sensor value exceeds a predefined threshold, a notification is sent to the user via the Blynk app; otherwise, the system continues monitoring.

4.4 HARDWARE IMPLEMENTATION

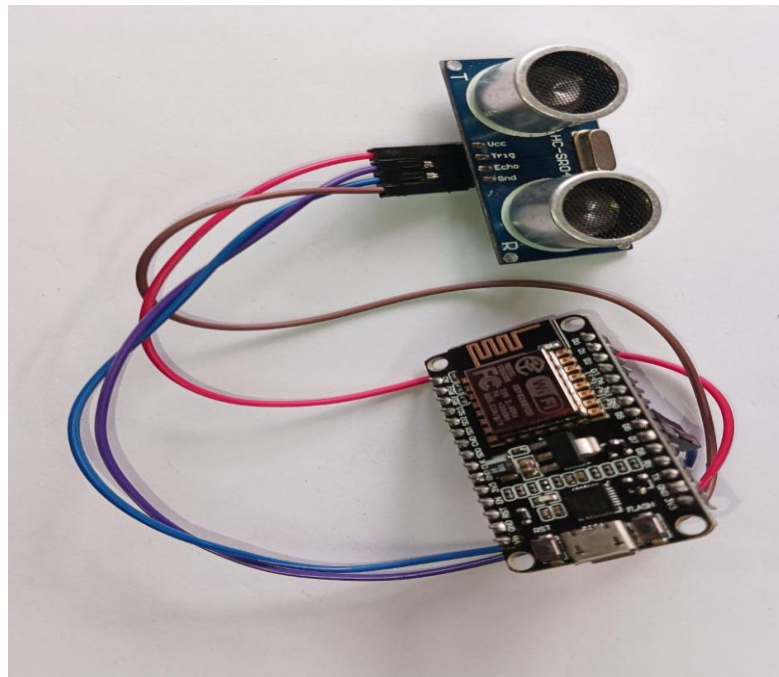


Figure 4.3 : Hardware Implementation of alerting system

1. Microcontroller – NodeMCU (ESP8266)

NodeMCU is the main controller that processes data from the ultrasonic sensor and connects to the internet via Wi-Fi. It has digital I/O pins for sensor connection and supports cloud communication through the Blynk platform.

2. Ultrasonic Sensor (HC-SR04)

Mounted at the top of the bin, this sensor continuously measures the distance to the garbage. It works on the principle of sound wave reflection and sends distance data to the NodeMCU.

3. Power Supply

The NodeMCU and ultrasonic sensor are powered using a 5V USB adapter, battery pack, or power bank. This ensures portability and stable operation.

4. Wi-Fi Connectivity

NodeMCU connects to the local Wi-Fi network to send real-time data to the Blynk cloud. This enables remote monitoring through the mobile app.

5.Blynk Mobile App

The app provides a user interface to monitor the bin's status, view distance readings, and receive notifications when the bin is full.

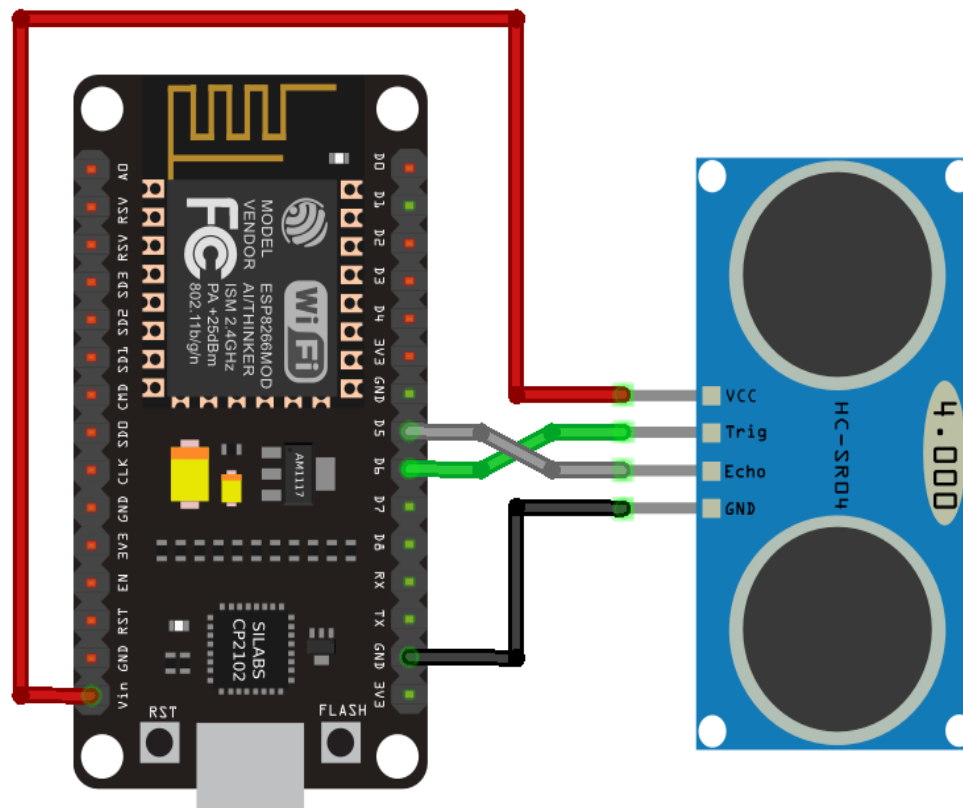


Figure 4.4 : Circuit diagram of Smart alerting system

4.5 WORKING OPEARTION

The working operation of the Smart Trash Alerting System begins with powering up the NodeMCU and establishing a Wi-Fi connection. The ultrasonic sensor, mounted at the top inside the dustbin, continuously emits ultrasonic waves and receives the echo reflected from the surface of the trash. The sensor measures the time taken for the echo to return and calculates the distance between the sensor and the trash level. This distance is then sent to the NodeMCU, which compares it with a predefined threshold value to determine whether the bin is empty, half-full, or full.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 INTRODUCTION

In this chapter we discussed about the results and applications, advantages and disadvantages of the project. This chapter presents the results obtained from the implementation of the Smart Trash Alerting System. It highlights how effectively the system detects bin levels and sends real-time alerts. The performance is analyzed based on parameters such as accuracy, response time, and reliability. Observations from testing are discussed to evaluate system efficiency. These results help in understanding the practical impact and areas for improvement. Overall, this chapter validates the functionality and effectiveness of the proposed system.

5.2 RESULTS

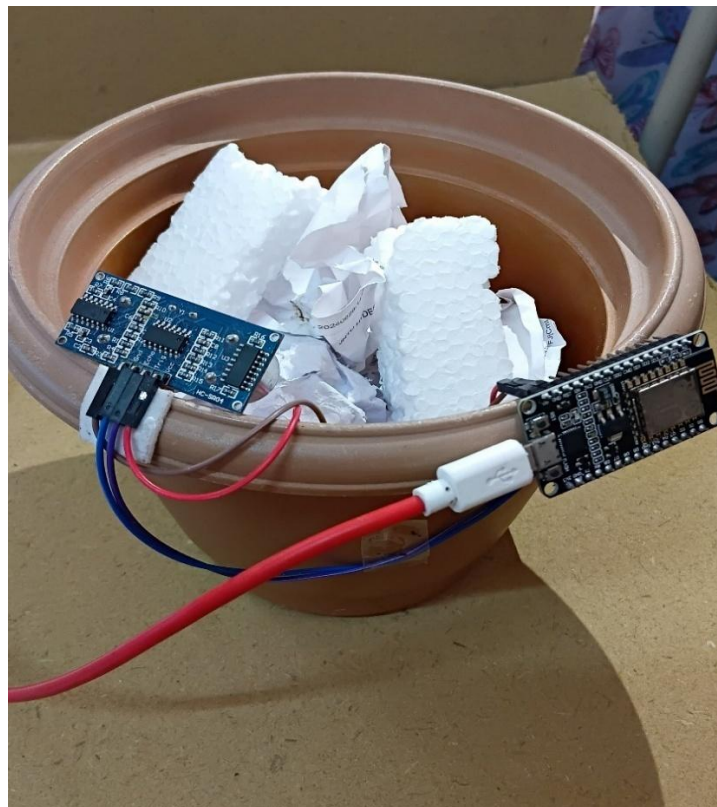


Figure 5.1 : IoT-Based Smart Dustbin



Figure 5.2 : Blynk App Notification

5.3 DISCUSSION

The Figure 5.2 shows the final output of the Smart Trash Alerting System using IoT, displayed through the Blynk app interface. It indicates that the trash bin has reached 90% of its capacity and provides the exact location using GPS coordinates. A pop-up alert is generated with a clear message prompting the user to empty the bin. The notification is a real-time response triggered by the ultrasonic sensor detecting the fill level. This result validates the system's successful integration of sensor data, cloud processing, and mobile alerts.

Advantages :

- Real-Time Monitoring
- Efficient Waste Management
- Remote Accessibility
- Cost-Effective Solution.
- Scalable Design

- Power-Efficient Operation
- Customizable Alerts
- User-Friendly
- Reduces Labor and Resource Waste
- Promotes Clean and Smart Cities

Disadvantages:

- Dependency on Internet Connectivity
- Security Vulnerabilities
- Maintenance Required

Applications:

1. **Smart Cities** – Automates waste monitoring in public areas.
2. **Offices and Workspaces** – Ensures timely disposal of waste in corporate environments.
3. **Hospitals** – Maintains hygiene by preventing bin overflows in medical facilities.
4. **Educational Institutions** – Helps in managing campus cleanliness efficiently.
5. **Shopping Malls and Complexes** – Monitors bin levels in large commercial spaces.
6. **Airports and Railway Stations** – Supports clean transport hubs with automated waste alerts.
7. **Residential Apartments** – Provides smart waste solutions for gated communities.
8. **Industrial Areas** – Manages waste in factories and production units effectively.
9. **Public Parks and Streets** – Prevents littering by alerting municipal staff on time.
10. **Event Venues** – Helps maintain cleanliness during large gatherings or festivals.

5.4 CONCLUSION

In this chapter we have discussed about the results of the project and we discussed about the advantages, disadvantages and applications of the project. The results confirm that the Smart Trash Alerting System functions accurately and efficiently in detecting bin fill levels. The system successfully generated real-time alerts, reducing the need for manual checks. These findings prove the feasibility and effectiveness of the proposed IoT-based solution.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The “**Smart Trash Alerting System using IoT**” offers a smart and reliable solution to monitor the status of bins in real time using affordable components like NodeMCU (ESP8266), an ultrasonic sensor, and the Blynk app. The system automatically detects the level of trash inside the bin and sends instant alerts to the user when it reaches a set limit. This eliminates the need for frequent manual checking and helps maintain a cleaner environment by ensuring that bins are emptied at the right time.

The project successfully combines hardware and software to provide a simple yet efficient monitoring system. The use of Wi-Fi communication enables users to receive notifications remotely, and the optional addition of a buzzer or LED gives an immediate on-site indication of bin status. The system has been designed with scalability and flexibility in mind, making it suitable for a variety of settings such as institutions, homes, public places, and offices. This project has been implemented, tested, and verified for accurate performance under normal conditions. It has proven to work consistently, providing real-time updates and clear notifications. All key objectives including trash level detection, alert generation, and mobile interface integration have been fully achieved.

In conclusion, the Smart Trash Alerting System is a completed and functioning prototype that effectively solves the problem of unnoticed bin overfill. It promotes cleaner surroundings through automation, and its successful implementation marks the end of the development phase. The project stands ready for real-world use or further improvement and scaling in the future.

6.2 FUTURE SCOPE

1. **Integration with Solar Power** : To improve sustainability, the system can be upgraded to operate using solar energy for interrupted, eco-friendly performance.
2. **Enhanced Mobile Application** : Additional features such as graphical data visualization, real-time bin mapping, and customized notifications can be added to the Blynk app.
3. **Multi-Bin Monitoring Capability**: The system can be expanded to track multiple bins simultaneously from a centralized interface, suitable for smart campuses or cities.

4. **GPS Tracking Support:** Adding GPS modules will help in real-time location tracking of bin, making it easier to manage bins spread across wide areas.
5. **Predictive Analytics With Machine Learning:** By implementing AI algorithms, the system can predict bin fill levels and recommend optimal collection schedules.
6. **Waste Classification Functionality :** Future versions can include smart sensors or image processing to differentiate between dry, wet, and hazardous waste.

REFERENCES

- [1] Raj, A., & Balakrishnan, V. (2019). IoT-based smart garbage monitoring system using NodeMCU. *International Journal of Engineering Research & Technology (IJERT)*, Vol. 8 Issue 7.
- [2] Sharma, P., & Gupta, R. (2020). Smart waste bin monitoring and alerting system using IoT. *International Journal of Scientific Research in Engineering and Management*, Vol. 4, Issue 2.
- [3] Kumar, V., & Singh, A. (2021). Real-time trash level monitoring using NodeMCU and ultrasonic sensor. *International Research Journal of Engineering and Technology (IRJET)*, Vol. 8, Issue 9.
- [4] Mahajan, S., & Patil, P. (2022). Smart bin system for urban cleanliness using IoT. *Journal of Emerging Technologies and Innovative Research (JETIR)*, Vol. 9, Issue 5.
- [5] Singh, S., & Mehta, R. (2023). IoT-enabled intelligent garbage bin for smart cities. *Journal of Science and Technology*, Vol. 12, Issue 3.
- [6] Bansode, S. S., & More, P. M. (2021). Design and implementation of smart dustbin using IoT for waste monitoring. *International Journal of Innovative Research in Science, Engineering and Technology (IJIRSET)*, Vol. 10, Issue 4.
- [7] Patel, R., & Shah, D. (2020). Smart IoT-based trash level detection and alert system using NodeMCU. *International Journal of Computer Applications (IJCA)*, Vol. 175, No. 25.